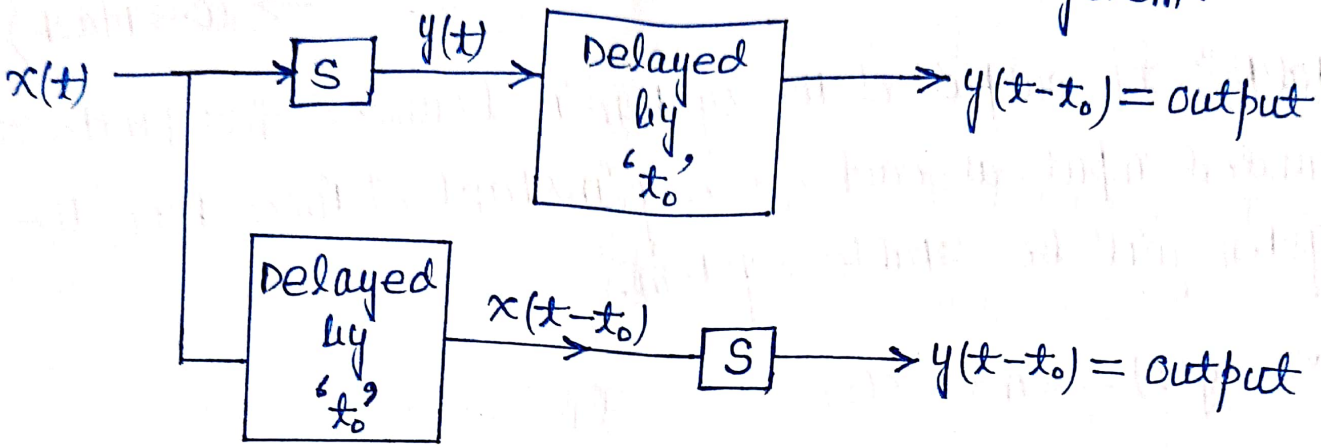
S4S QUIZ-2 SOLUTION

Solution 1(a):-

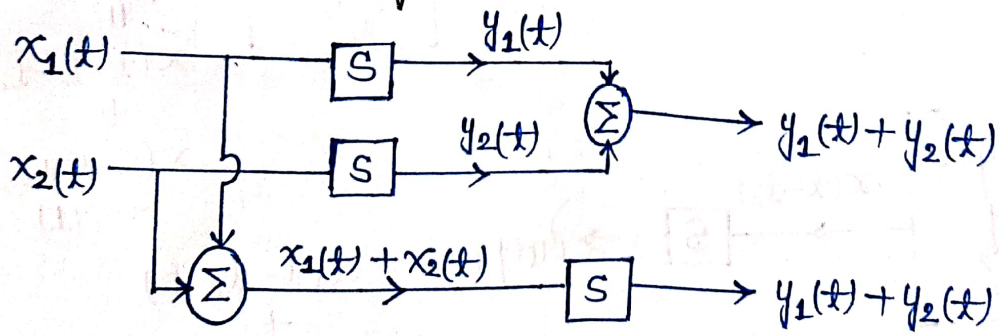
Time invariant :- Any delay provided at input must be reflected in output for Time invariant system.



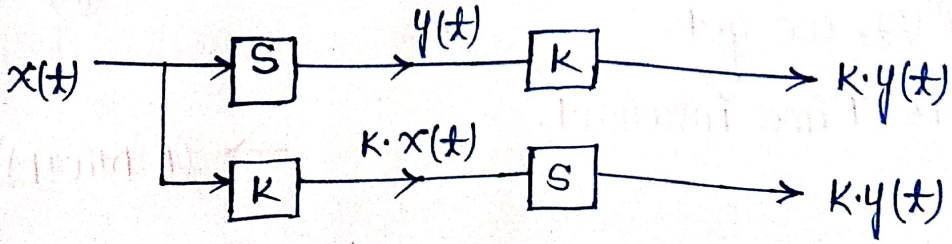
Ex: $y(t) = x(\sin t)$ → (0.5 Mark)

Linear :- A linear system follows the law of superposition. It is a combination of two laws — (I) Law of additivity (II) Law of Homogenity Ex: $y(t) = \cos t \cdot x(t)$

(I) **Law of additivity**



(II) **Law of Homogenity**



→ (0.5 Mark)

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Causal:- If output of the system is independent of future value of input at each & every instant of time then system will be causal system.

Ex: $y(t) = x(t) + x(t-1)$

→ (0.5 Mark)

Stable:- If output of the system is bounded for finite or bounded input at each & every instant of time then the system will be stable system.

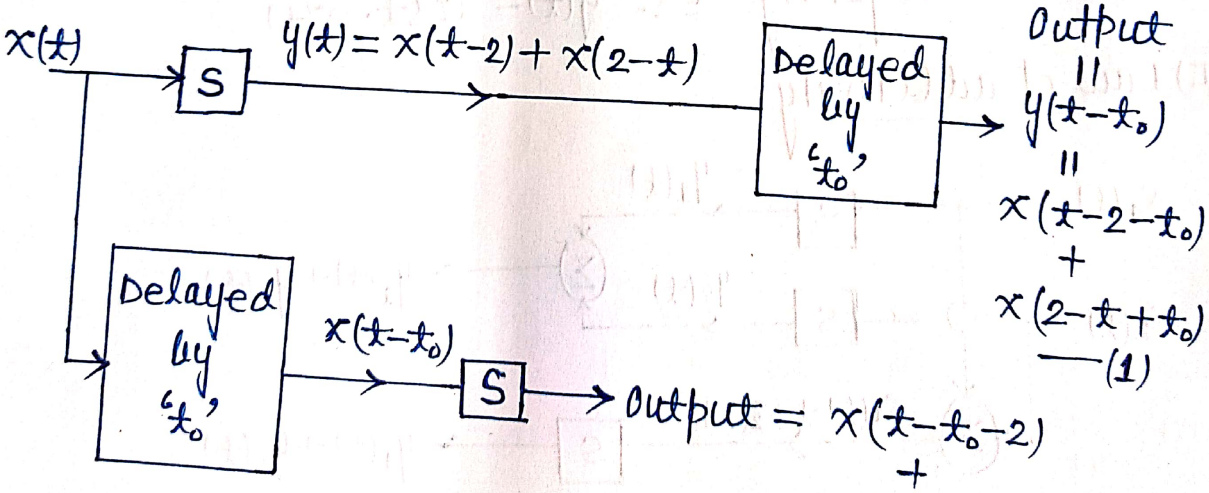
Ex: $y(t) = \sin t \cdot x(t)$

→ (0.5 Mark)

Solution 1(b):-

(I) Given— $y(t) = x(t-2) + x(2-t)$

Step(1):-



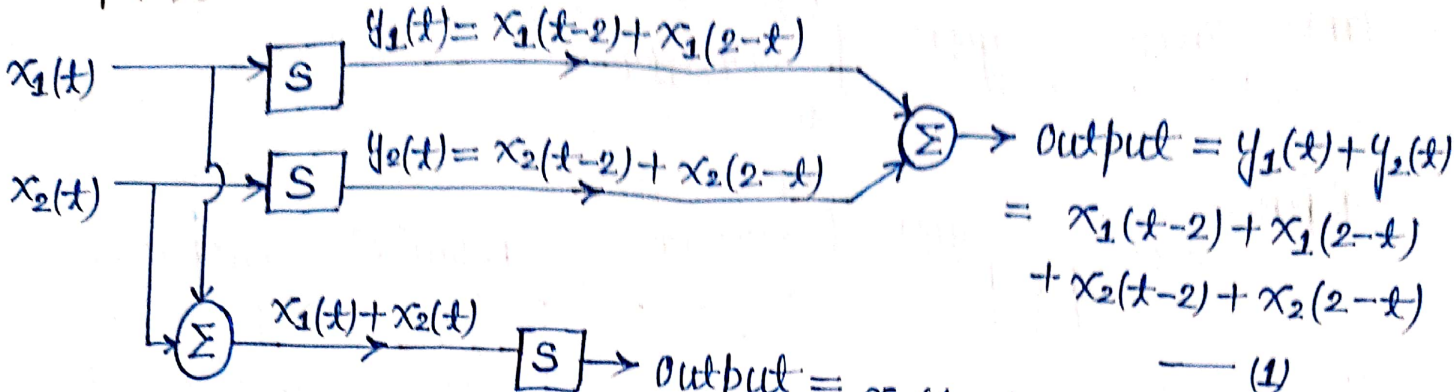
By eqn(1) & eqn(2), we get-

System is Time invariant.

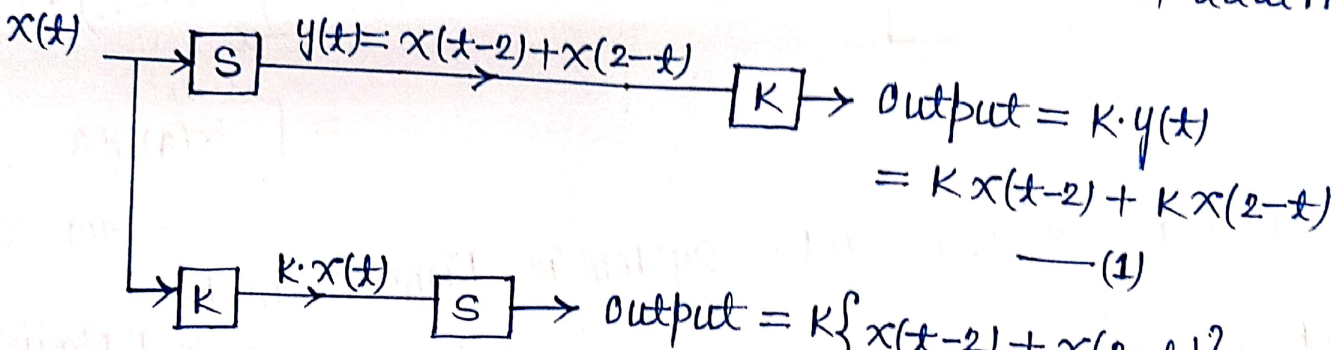
→ (1 Mark)

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Step (II):-



By eqⁿ (1) & eqⁿ (2), we get – system follow law of additivity



By eqⁿ (1) & eqⁿ (2), we get – System follow law of Homogenity.

Hence, system is Linear. → (1 Mark)

Step (3):- Let's check at, $t = 0$

$$y(t) = x(+2) + x(-2)$$

(Future Value, Past Value)

Hence, system is Non-Causal. → (1 Mark)

Step (4):- System is Stable → (1 Mark)

Let, $|x(t)| \leq M$

$\therefore |y(t)| = |x(t-2) + x(2-t)| \leq 2M$

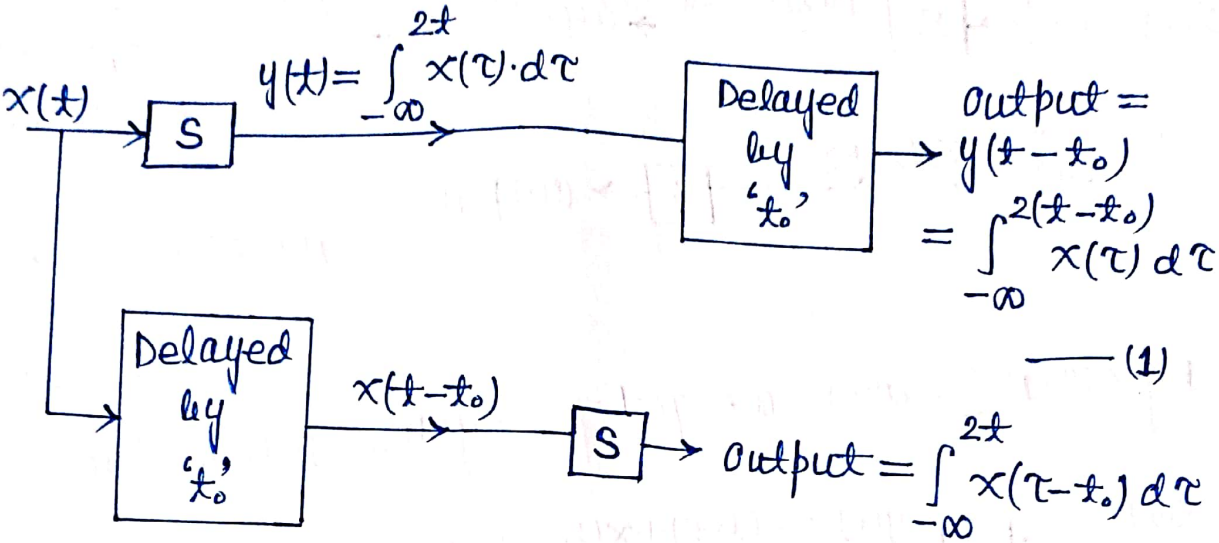
Output will always a finite value.

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Solution 1(b):-

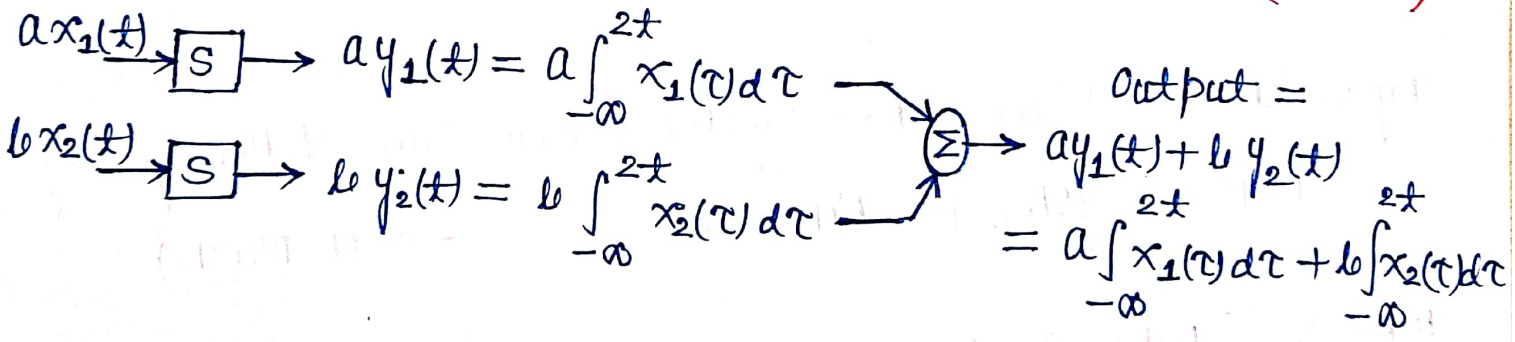
(II) Given- $y(t) = \int_{-\infty}^{2t} x(\tau)\cdot d\tau$

Step(1):-



By eqⁿ(1) & eqⁿ(2), we get- System is Time Variant.

Step(2):- → (1 Mark)



$ax_1(t) + bx_2(t)$ → S → Output = $\int_{-\infty}^{2t} \{ax_1(\tau) + bx_2(\tau)\}\, d\tau$ —— (2)

By eqⁿ(1) & eqⁿ(2), we get- System is Linear. → (1 Mark)

Step(3):- At time 't', upper limit is '2t', so it depends on future value of x(t) if $2t > t$

Hence, System is Non-causal. → (1 Mark)

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Step (4):- Let's assume, $|x(t)| \leq M$ then -

$$|y(t)| = \left| \int_{-\infty}^{2t} x(\tau)\, d\tau \right| \leq \int_{-\infty}^{2t} |x(\tau)|\, d\tau$$

This integration over an infinite interval, which can diverge. So even if input is bounded, the output can be unbounded.

Hence, system is Not-Stable. → (1 Mark)

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